# Infix to Postfix Expression Conversion Algorithm Using Stack

Steps:

1. Scan Infix expression from Left to Right.

2. If current character is operand
   Then add it to output expression (postfix).

3. If current character = '('
   Then push it onto Stack.

4. If current character = ')'
   Then
   - 4.1 Pop Stack Top and add it to output expression.
   - 4.2 Repeat step 4.1 until a matching ')' is found.
   - 4.3 Pop matching ')' of Stack and ignore it.

5. If current character is operator
   Then
   - 5.1 If Stack is empty, Then push current operator onto Stack.
   - 5.2 If Stack Top = '(', Then push current operator onto Stack.
   - 5.3 If precedence of current operator **>** precedence of Stack Top,
     Then push current operator onto Stack.
     Else pop Stack Top and add it to output expression.
   - 5.4 Repeat step 5.3 until current operator is pushed onto Stack.

6. If End of Infix expression,
   Then pop (dump) Stack content and add it to output expression.